

Section A:

1. What is SQL? Why is it important in DBMS?

Ans: SQL (Structured Query Language) is a standard language used to interact with relational databases. It is used to create tables, insert data, update records, delete data, & retrieve information.

Importance:

- Acts as a bridge between users & databases.
- Enables data retrieval & manipulation.

2. Difference between DDL & DML.

DDL (Data Definition Language)

- Defines database structure.
- Commands :- CREATE, DROP, ALTER
- Affects schema

DML (Data Manipulation Language)

- Manipulates data inside tables.
- Commands :- INSERT, UPDATE, DELETE.
- Affects data

3. What is a JOIN? Explain its types:

Ans: JOIN combines data from multiple tables.

Types:

- 1) INNER JOIN: Returns matching records from both tables.
- 2) LEFT JOIN: Returns all records from left table & matched from right.
- 3) RIGHT JOIN: Returns all records from right table & matched from left.
- 4) FULL JOIN: Returns all records from both tables (matched & unmatched)

4) What is?

- GROUP BY: Groups rows based on a column (used with aggregate functions like COUNT, SUM)
- HAVING: Filters grouped results (used after GROUP BY).

5. What is subquery?

Ans: A query written inside another query.

6. What is relational algebra?

Ans: It is a procedural query language used in DBMS. It defines a set of operations (like select, join) to retrieve data from RD.

7. Difference betn WHERE & HAVING?

WHERE

Filters rows before grouping
Cannot use aggregate function
Applied on individual rows

HAVING

Filters groups after Group By.
Can use aggregate functions.
Applied on grouped data.

Section B: Design, Write & Defend:

1. Retrieve students with marks > 75

Select * from students where marks > 75;

2. JOIN query (Student + Course)

Select student.name, Course.course-name from student INNER JOIN Course ON student.Course-id = Course.course-id;

3. Count students per department

Select department, COUNT(*) As total_students from students, GROUP BY department;

Section C:

1. Employees in a specific department

Select Employee.name from Employee INNER JOIN Department ON Employee.dept-id = Department.dept-id Where department.dept-id = "HR";

2. Total sales per customer

Select customer-id, SUM (amount) As total-sales from orders group by customer-id;

3. Top 3 students:

select * from students order by marks desc limit 3;

Section : 2

1. Real-Life SQL Query + Logic

→ select name, salary from employee where salary > 50000;

→ Logic: Retrieves employees earning more than 50,000, used for identifying high-paid staff.

2. JOIN Query + use case

→ select Orders.order_id, Customer.name from orders INNER JOIN Customer on Orders.customer_id = Customer.customer_id;

→ Use: Used in e-commerce to display which customer placed which order.

3. GROUP BY query + explanation:

→ select department, Avg(salary) as avg_salary from employees GROUP BY department;

→ Explanation: Groups employees by department & calculates average salary for each group.

☑ VIVA Component:

1. Why SQL important?

Manages & retrieves data efficiently.

2. Incorrect JOIN?

Wrong results or large unwanted data

3. GROUP BY example?

Total orders per customer.

4. Complex queries impact?

Slower performance.